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New Sampling Formulae Associated with the Linear Canonical Transform

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Abstract

Linear canonical transform play an important role in many fields of optics and signal processing society. Well known transforms in these fields such as the Fourier transform, the fractional Fourier transform, and the Fresnel transform can be looked at as special cases of the linear canonical transform. In this paper we obtain new sampling formulae for reconstructing signals that are band-limited or time-limited in the linear canonical transform sense. In the new sampling formulae, we use samples from both the signal and its Hilbert transform, but each taken at half the Nyquist rate of linear canonical transform domain.

1.Introduction

The linear canonical transform (LCT)[1]-[3] is a powerful tool for optics processing, radar system analysis, filter design, phase retrieval, and many other applications [2]. It is an integral transform with four parameters $\{a,b,c,d\}$. Many useful transforms, such as the Fourier transform (FT), fractional Fourier transform (FrFT)[3]-[5], Fresnel transform (FrT)[6], and scaling operations are all special cases of the LCT[2] .

At the same time, the sampling process is a fundamental work converting the continuous signals to the discrete signals; it is central in almost all domain because it provides the links between the continuous physical signals and the discrete domain. The most common sampling theorem is the Shannon sampling theorem[7] which states that if a function $f(t)$ contains no frequencies higher than π / T rad/s, it can be perfectly reconstructed provided the sample space is less than T s. The sampling expansion for the linear canonical transform of compact functions has been derived recently in [8]-[9], the formulae they derived can be used to reconstruct the original compact signals or its linear canonical transform from their samples at a discrete set of points satisfying the condition of sampling theorems of that domain.

However, in some real applications, when sampling broadband signals or non-stationary signals, the sampling rate does not satisfy the condition of sampling theorem of that domain [10],[11]. So it is worthwhile and interesting to obtain new sampling formulae that can keep the sampling rate as low as possible.

Based on the newly proposed sampling theorem of linear canonical transformed signals[8],[9], we obtained new sampling formulae in this paper for reconstructing signals that are band-limited or time-limited in the linear canonical transform sense. In the new sampling formulae, we use samples from both the signal and its Hilbert transform, but the sampling rate can be taken at half the sampling rate of [8], [9].

The rest of the paper is organized as follows. Section 2 provides a brief review of linear canonical transform and its properties. In section 3, a new sampling formula associated with the linear canonical transform is presented in detail. Finally, in section 4 the conclusions and future works along this way are given.

2.Preliminaries

2.1 The linear canonical transform

The linear canonical transform of a signal $f(t)$ with

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parameters $\mathbf{A}=(a,b,c,d)$ is defined as [2]:

$$L_f^{\mathbf{A}}(u)=L_f^{\mathbf{A}}\{f(t)\}(u)$$

$$=\begin{cases}\sqrt{\dfrac{1}{j2\pi b}}\displaystyle\int_{-\infty}^{+\infty} f(t)e^{j\frac{1}{2}[\frac{a}{b}t^2-(\frac{2}{b})tu+\frac{d}{b}u^2]}dt, & b\neq 0,\\ \sqrt{d}e^{j\frac{1}{2}cdu^2}f(du), & b=0,\end{cases} \quad (1)$$

Where a,b,c,d are real numbers and satisfying $ad-bc=1$.

It is easy to verify that the LCT with parameters $\{a,b,c,d\}=\{\cos\theta,\sin\theta,-\sin\theta,\cos\theta\}$ reduces to the FrFT[3]-[5] which, in the specific case $\theta=\pi/2$, becomes the FT. With parameters $\{a,b,c,d\}=\{1,b,0,1\}$ the LCT reduces to the FrT. Scaling operation can be viewed as a special case of the LCT with parameters $\{a,b,c,d\}=\{d^{-1},0,0,d\}$ [2].

Therefore, the FT, FrFT and scaling operations are all special cases of the LCT. And the results in these transform domains can be extended to the LCT domain, in this sense the LCT can extend their utilities and applications and additionally may solve some problems that cannot be solved well by these operations [8]. For more properties and applications of LCT in signal and optics processing one can refer to [2],[3].

A signal $f(t)$ is called $\Omega_{\mathbf{A}}$ band-limited signal in linear canonical transform sense, which means that

$$L_f^{\mathbf{A}}(u)=L_f^{\mathbf{A}}\{f(t)\}(u)=0 \quad \text{For } |u|>\Omega_{\mathbf{A}} \qquad (2)$$

Where $\Omega_{\mathbf{A}}$ is called the bandwidth of signal $f(t)$ in the linear canonical transform domain. The sampling theorem expansion of band-limited signal in the linear canonical transform domain can be represented as following [8], [9].

Lemma 1. Let $f(t)$ be a signal band limited to $\Omega_{\mathbf{A}}$ (where $\mathbf{A}=\{a,b,c,d\}$) in the LCT sense, i.e., the support of $L_f^{\mathbf{A}}(u)$ is $[-\Omega_{\mathbf{A}},\Omega_{\mathbf{A}}]$. Then the original signal $f(t)$ can be exactly reconstructed from its sampled version $f(nT)$, $n\in Z$

$$f(t)=e^{-j\frac{a}{2b}t^2}\sum_{n=-\infty}^{+\infty} f(nT)e^{j\frac{a}{2b}(nT)^2}\cdot\frac{\sin[\Omega_{\mathrm{A}}(t-nT)/b]}{\Omega_{\mathrm{A}}(t-nT)/b} \qquad (3)$$

Where $T=\dfrac{1}{f}=\pi b/\Omega_{\mathbf{A}}$, and f is the Nyquist rate of signal associated with LCT.

2.2 The Hilbert transform

The Hilbert transform is also known to play an important role in signal processing and optics analysis. The Hilbert transform of a signal $f(t)$ is defined as :

$$H\{f(t)\}(y)=\frac{1}{\pi}\int_{-\infty}^{+\infty}\frac{f(t)}{y-t}dt \qquad (4)$$

The analytic part of $f(t)$ is defined through the Hilbert transform of the signal as:

$$F(t)=f(t)+jH[f](t) \qquad (5)$$

One of the most important properties of analytic signals is that they contain no negative Fourier frequency components. This fact is used in the following sections to derive sampling theorem expansions for band-limited signals using samples from both the signal and its Hilbert transform, but each taken at half the Nyquist rate in linear canonical transform domain.

3.The new sampling formulae

In this section, we derive the new sampling formulae associated with the linear canonical transform. And we present the main results of this paper as a theorem.

Theorem 1. Let $f(t)$ be a real signal and suppose that $e^{-jat^2/2b}f(t)$ is band limited to $[-\Omega_{\mathbf{A}},\Omega_{\mathbf{A}}]$ in the sense of the LCT. Then the following sampling expansion for $f(t)$ holds:

$$f(t)=\sum_{n=-\infty}^{+\infty}\{f(nT_1)\cos[\frac{\Omega_A}{2b}(t-nT_1)]-H[f](nT_1)\cdot\sin[\frac{\Omega_A}{2b}(t-nT_1)]\}\frac{\sin[\Omega_A(t-nT_1)/2b]}{\Omega_A(t-nT_1)/2b} \quad (6)$$

where $T_1=\dfrac{1}{f_1}=2\pi b/\Omega_{\mathbf{A}}$.

Proof: let $g(t)$ be the analytic signal of $f(t)$, i.e., $g(t)=f(t)+jH\{f\}(t)$, multiplying by $e^{-jat^2/2b}$ and taking the LCT of both sides with parameters $\mathbf{A}$=(a,b,c,d), we have

$$L_g^{\mathbf{A}}\{e^{-jat^2/2b}g(t)\}(u)$$

$$=L_f^{\mathbf{A}}\{e^{-jat^2/2b}f(t)\}(u)+jL_{Hf}^{\mathbf{A}}\{e^{-jat^2/2b}H[f](t)\}(u) \quad (7)$$

Andbecause

$$L_{Hf}^{\mathbf{A}}\{e^{-jat^2/2b}H[f](t)\}(u)$$

$$=\sqrt{\frac{1}{j2\pi b}}e^{j\frac{d}{2b}u^2}\int_{-\infty}^{+\infty}f(x)(\frac{1}{\pi}\int_{-\infty}^{+\infty}\frac{e^{-j\frac{1}{b}tu}}{t-x}dt)dx$$

$$=\sqrt{\frac{1}{j2\pi b}}e^{j\frac{d}{2b}u^2}\int_{-\infty}^{+\infty}f(x)H[e^{-j\frac{1}{b}tu}](x)dx$$

Where $H[e^{-jtu/b}](x)$ denotes the Hilbert transform of signal $e^{-jtu/b}$ andn view of the fact that

$$H[e^{-jtu/b}](x)=-j\,\mathrm{sgn}(u)e^{-j\frac{1}{b}xu} \text{ for } b>0$$

Thus we have

$$L_{Hf}^{\mathbf{A}}\{e^{-jat^2/2b}H[f](t)\}(u)$$

$$=(-j)\mathrm{sgn}(u)\sqrt{\frac{1}{j2\pi b}}e^{j\frac{d}{2b}u^2}\int_{-\infty}^{+\infty}f(x)e^{-j\frac{1}{b}xu}dx$$

$$=(-j)\mathrm{sgn}(u)L_f^{\mathbf{A}}[e^{-j\frac{a}{2b}x^2}f(x)](u)$$

Therefore, by substituting the above equation into equation (7), we obtain:

$$L_g^{\mathbf{A}}\{e^{-jat^2/2b}g(t)\}(u)=(1+\mathrm{sgn}(u))L_f^{\mathbf{A}}[e^{-j\frac{a}{2b}t^2}f(t)](u) \quad (8)$$

Equation (8) implies that $L_g^{\mathbf{A}}\{e^{-jat^2/2b}g(t)\}(u)$ supported in $[0,\Omega_A]$, if we denote $e^{-jat^2/2b}g(t)$ by $h(t)$; it then follows from lemma 1 that $h(t)$ is a compact signal in linear canonical transform sense, and it can be looked at as band-passed to $[\frac{\Omega_A}{2}-\frac{\Omega_A}{2},\ \frac{\Omega_A}{2}+\frac{\Omega_A}{2}]$. Following the results in [10] we exactly reconstructed the signal $h(t)$ from its sampled version $h(nT)$, $n\in Z$ by the reconstruction formula:

$$h(t)=e^{-j\frac{a}{2b}t^2}\sum_{n=-\infty}^{+\infty}h(nT_1)e^{j\frac{a}{2b}(nT_1)^2}\cdot\frac{\sin[\Omega_A(t-nT_1)/2b]}{\Omega_A(t-nT_1)/2b}\cdot e^{j[(\frac{\Omega_A}{2b})(t-nT_1)]} \quad (9)$$

Where $T_1=2\pi b/\Omega_{\mathbf{A}}$

So the original signal $g(t)$ can be rewritten as:

$$g(t)=e^{jat^2/2b}h(t)$$

$$=\sum_{n=-\infty}^{+\infty}g(nT_1)\frac{\sin[\Omega_A(t-nT_1)/2b]}{\Omega_A(t-nT_1)/2b}e^{j[(\frac{\Omega_A}{2b})(t-nT_1)]}$$

From the definition of the signal $g(t)$ we know that the original signal $f(t)$ is the real part of $g(t)$, and $f(t)$ can be derived as:

$$f(t)=\sum_{n=-\infty}^{+\infty}\{f(nT_1)\cos[\frac{\Omega_A}{2b}(t-nT_1)]-H[f](nT_1)\cdot\sin[\frac{\Omega_A}{2b}(t-nT_1)]\}\frac{\sin[\Omega_A(t-nT_1)/2b]}{\Omega_A(t-nT_1)/2b}$$

and which proves the theorem.

By interchanging the roles of $f(t)$ and its linear canonical transform $L_f^{A}(u)$, we obtain the following corollary easily.

Corollary 1. If $f(t)$ is time limited to $[-\Omega_A, \Omega_A]$. Then the following sampling expansion for linear canonical transform $L_f^A(u)$ holds:

$$L_f^A(u) = \sum_{n=-\infty}^{+\infty} \{L_f^A(nT_1)\cos[\frac{\Omega_A}{2b}(t-nT_1)] - H[L_f^A](nT_1) \cdot \sin[\frac{\Omega_A}{2b}(t-nT_1)]\} \frac{\sin[\Omega_A(t-nT_1)/2b]}{\Omega_A(t-nT_1)/2b} \quad (10)$$

Corollary 2. When the linear canonical transform parameter reduces to $\mathbf{A} = \{\cos\theta, \sin\theta, -\sin\theta, \cos\theta\}$, the results reduce to the classical results of fractional Fourier transform domain proposed in [12] which, in the specific case $\mathbf{A} = \{0, 1, -1, 0\}$, becomes the classical results in Fourier transform domain.

It is shown in this sense that the classical results in Fourier transform domain[11] or fractional Fourier transform domain[12] are all special cases of our results.

4.Conclusions and future works

Two new sampling formulae for reconstructing signals that are band-limited or time-limited in the linear canonical transform sense are presented. In the new sampling formulae, we use samples from both the signal and its Hilbert transform, but each taken at half the Nyquist rate of linear canonical transform domain. The well known sampling theorems in fractional Fourier domain or Fourier domain are all special cases of the newly developed formula. It is applicable for a larger class of cases, especially for cases that we have the samples of signal and its Hilbert transform.

The future works along this way is to deduce and obtain the new interpolation formulae for non-uniformly sampled signals or high-dimensional signals and the applications of these results in real situations.

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